

WASP-2b

R-1369

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_140825 · created 2026-08-01T00:33:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456894.8691 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.047
Transit depth (Rp/Rs)^2	2.36 +/- 1.45 [%]
Orbital Inclination (inc)	87.15 +/- 2.8
Airmass coefficient 1 (a1)	1.25 +/- 0.41
Airmass coefficient 2 (a2)	-0.151 +/- 0.046
Scatter in the residuals of the lightcurve fit is	30.92 %
Best Comparison Star	None
Optimal Aperture	3.4
Optimal Annulus	3.86
Transit Duration (day)	0.085 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.047 vs 0.1326 (deviation 0.46σ)
Duration fit vs published	0.085 d vs 0.0737 d (deviation 0.54σ)
Mid-time O–C	9.0 min
Depth SNR	3.3
Residual scatter	30.92 %

Light curve

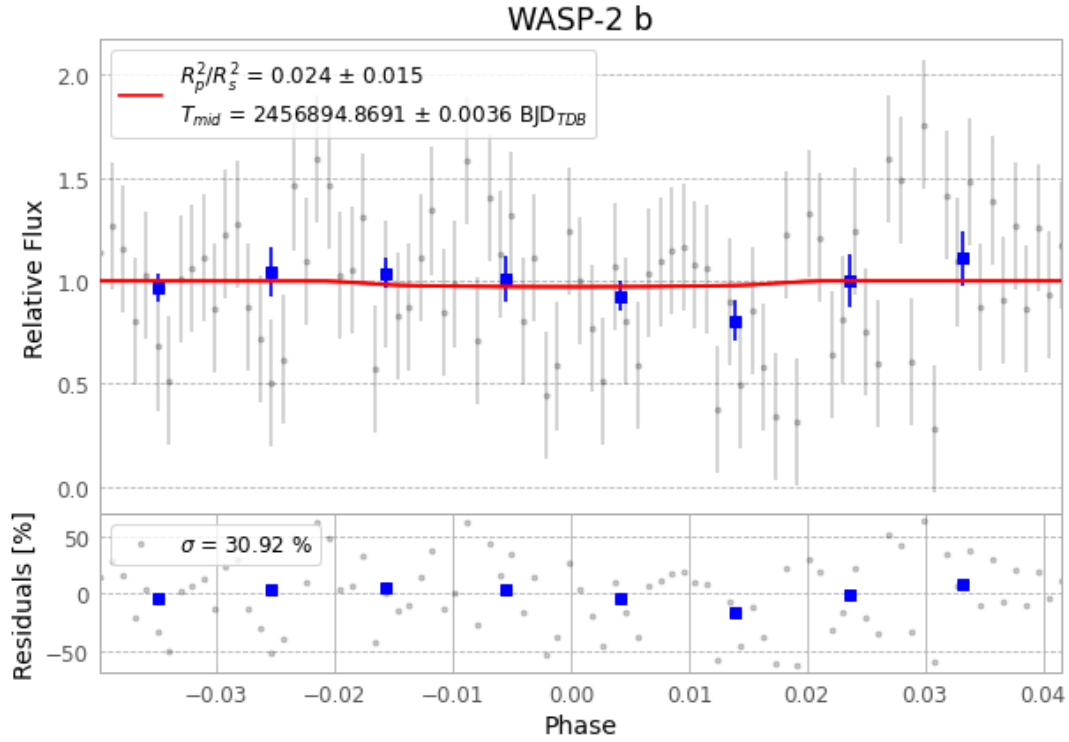


Figure 1 — FinalLightCurve_WASP-2 b_2014-08-24.png (SHA-256 b34360e608dedf11...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 240f64ef1a3a025c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), WASP-2140825063912.FITS → WASP-2140825105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-140825.log (a6ca18b85876...), FinalLightCurve_WASP-2 b_2014-08-24.png (b34360e608de...), FinalLightCurve_WASP-2 b_2014-08-24.csv (cb0ad8e5c3bb...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:33Z.